

E-COMMERCE CONVERSION DYNAMICS: CUSTOMER BEHAVIOR & PURCHASE INTENT ANALYSIS

Customer Behavior, Conversion
Dynamics & Purchase Intent



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly &
ML libraries

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Overview & Business Objective



1.1 PROJECT OVERVIEW

Retail businesses operate in a dynamic environment where demand is influenced by factors such as historical sales, inventory, pricing, promotions, seasonality, and region.

Accurate demand estimation helps businesses maintain optimal inventory, reduce stockouts and excess stock, and improve procurement and operational planning.

The Retail Demand Intelligence project uses historical retail data and machine learning to predict demand and identify the key factors influencing it.

The project combines:

- Data understanding and cleaning
- Exploratory Data Analysis
- Feature engineering
- Machine learning regression
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented insights

The objective is not only to predict demand accurately, but also to understand why the model makes its predictions & how these insights can support better decision-making.



1.2 PROBLEM STATEMENT



Retail demand is influenced by multiple interacting variables, making manual forecasting or reliance on historical averages insufficient for many situations.

The key problem addressed in this project is:

How can historical retail data be used to build a machine learning model that accurately predicts product demand while also identifying the major factors driving demand?

The project therefore treats Demand as the target variable and uses historical, operational, pricing, promotional, environmental, and temporal information as predictive features.



1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Develop a machine learning-based retail demand prediction system.
- Understand the underlying structure and characteristics of the dataset.
- Identify important patterns through exploratory data analysis.
- Create meaningful temporal, inventory, pricing, and historical features.
- Train and compare multiple regression models.
- Evaluate model performance using appropriate regression metrics.
- Investigate the largest prediction errors.
- Identify the most influential demand-driving features.
- Identify limitations and possible improvements for future development.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, demand modeling, model explainability, and business interpretation to understand demand patterns, evaluate predictive performance, identify key demand drivers, and derive actionable retail insights.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Distribution analysis
- Relationship analysis

02

Demand Modeling

- Feature preparation
- Regression modeling
- Model comparison
- Model evaluation

03

Explainability

- Feature importance analysis
- Original-feature aggregation
- Investigation of large prediction errors

04

Business Interpretation

- Business Interpretation
- Inventory implications
- Pricing and promotion observations
- Seasonal and weather-related effects
- Recommendations for retail planning



Data Understanding & Exploratory Data Analysis



2.1 DATA OVERVIEW

The dataset provides a comprehensive view of retail operations across stores and products, covering inventory levels, sales activity, ordering patterns, pricing, discounts, promotions, weather conditions, seasonality, and demand. These variables provide the foundation for analyzing demand behavior, identifying operational patterns, and developing predictive models for retail demand.

Data Source: Kaggle

Dataset Size

76,000

Records

16

Original Features

Purchase Diversity

4

Regions

5

Stores

20

Products



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Date	Date / Time	Date of the observation
Store ID	Categorical	Unique identifier of the store
Product ID	Categorical	Unique identifier of the product
Category	Categorical	Product category
Region	Categorical	Geographic region of the store
Inventory Level	Numerical (Discrete)	Inventory available at the time of observation
Units Sold	Numerical (Discrete)	Number of units sold
Units Ordered	Numerical (Discrete)	Number of units ordered
Price	Numerical (Continuous)	Selling price of the product
Discount	Numerical (Discrete)	Discount applied to the product
Weather Condition	Categorical	Weather condition during the observation
Promotion	Binary	Indicates whether a promotion was active
Competitor Pricing	Numerical (Continuous)	Price of the comparable product offered by competitors
Seasonality	Categorical	Seasonal classification of the observation
Epidemic	Binary	Indicates whether an epidemic-related condition was present
Demand	Numerical (Discrete)	Target variable representing product demand

2.2 TARGET VARIABLE

The target variable selected for prediction is:

Demand

Demand represents the quantity that the model attempts to estimate based on the available information

The target has:

104.32

Mean

100

Median

4

Minimum

430

Maximum

46.96

Standard Deviation

The distribution is concentrated around approximately 70-130 units, with a noticeable right tail extending toward higher demand values.



2.3 DATA QUALITY & PREPARATION OVERVIEW

Before modeling, the dataset was examined for:

- Data types
- Missing values
- Duplicate observations
- Categorical variables
- Numerical variables
- Temporal information
- Potential outliers
- Feature relationships

The *Date* variable was converted into usable temporal components, while categorical variables were prepared for machine learning using one-hot encoding.





2.4 Exploratory Data Analysis

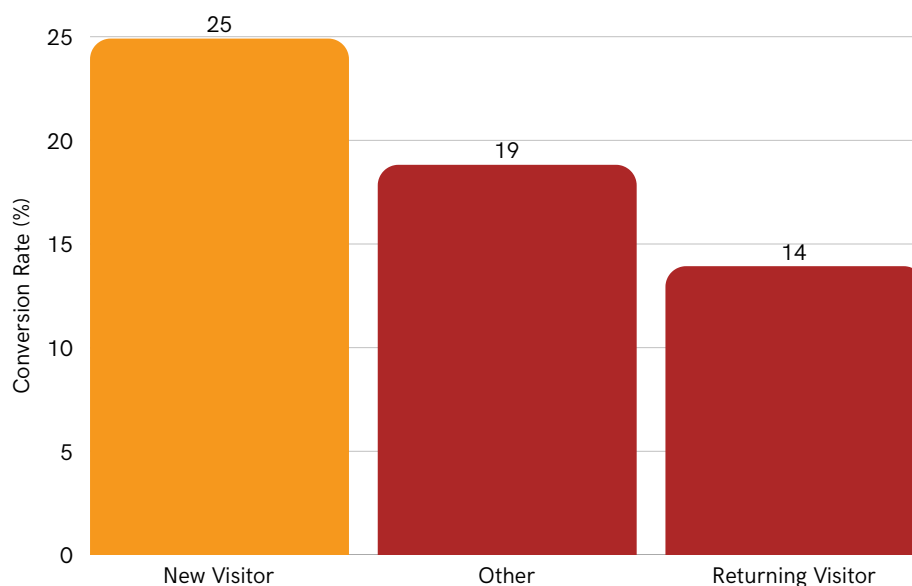


New visitors show the highest purchase conversion rate, while returning visitors record the lowest.

\$2,135

Average Cost per Service

Purchase Conversion Rates Across Traffic Sources



Key observations

- New visitors have the highest conversion rate at 24.91%, followed by the Other visitor group at 18.82%.
- Returning visitors have the lowest conversion rate among the three groups at 13.93%.

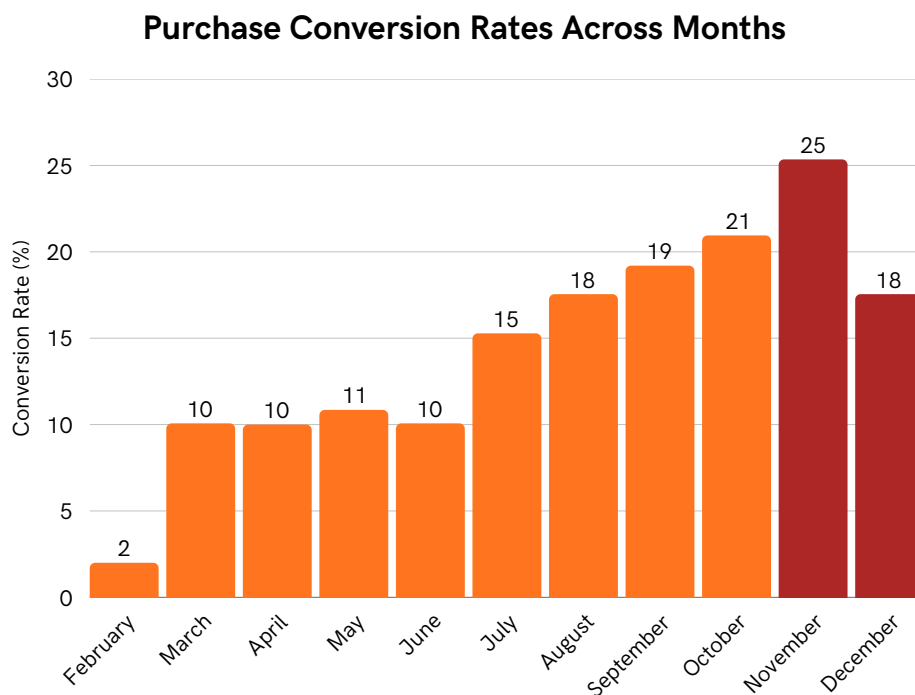
Business Insights

- Examine the browsing and acquisition characteristics of new visitors to understand the behaviors associated with their higher conversion rate.
- Investigate the lower conversion rate among returning visitors to identify potential opportunities to improve repeat-session conversion.

Purchase conversion rates vary substantially across months, with November showing the highest conversion rate.

\$1,251

Typical Cost per Resource (Median)



Key observations

- November records the highest conversion rate at 25.35%, followed by October at 20.95%.
- February records the lowest conversion rate at 1.63%, while March to June records around 10%.

Business Insights

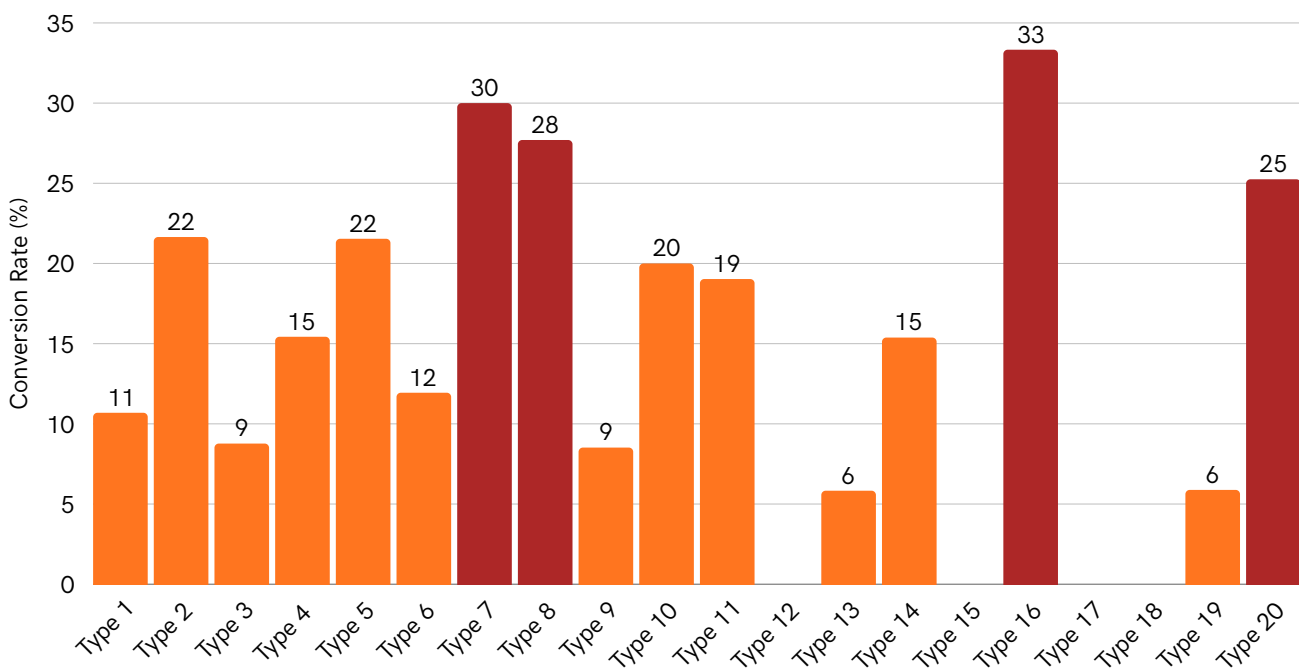
- Prioritize conversion-focused campaigns and promotional activity during months showing stronger conversion rates, particularly around October and November.
- Investigate the behavioral and commercial factors associated with low-conversion months such as February to identify opportunities for improvement.

Purchase conversion rates vary considerably across traffic sources, with a small number of traffic types showing substantially higher conversion rates.

\$2,147

Average Spending per Region

Purchase Conversion Rates Across Traffic Type



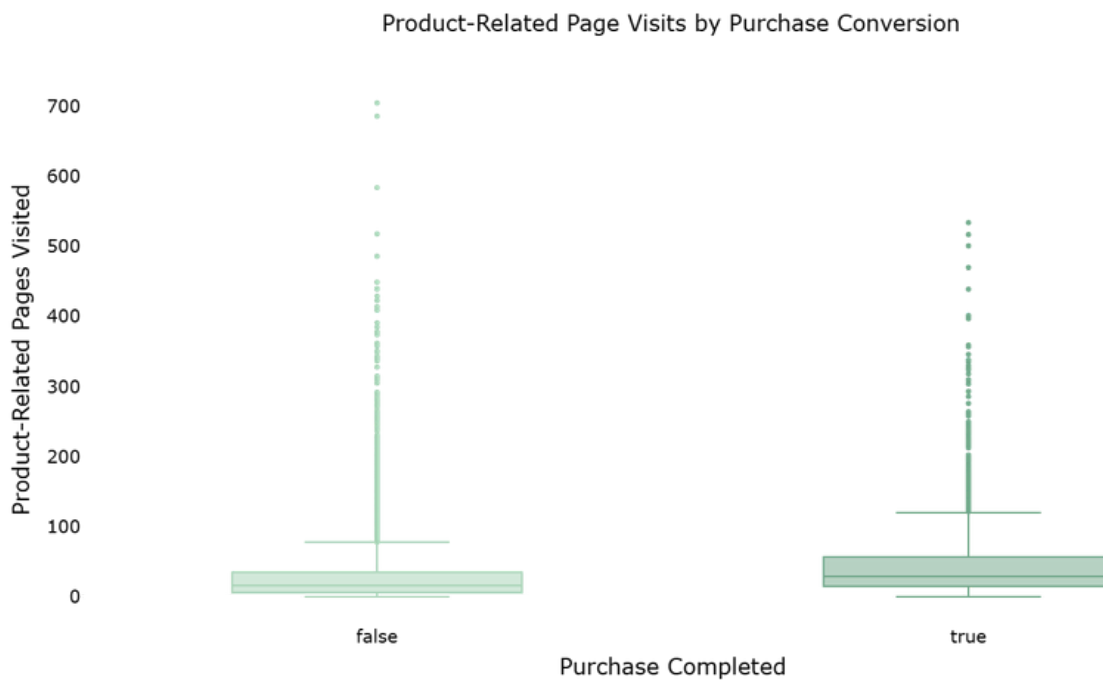
Key observations

- Traffic Type 16 records the highest conversion rate at 33.33%, followed by Traffic Type 7 at 30.00% and Traffic Type 8 at 27.70%.
- Several traffic types record very low or 0% conversion rates, while Traffic Type 20 reaches 25.25%.

Business Insights

- Evaluate high-conversion traffic sources to understand which acquisition channels are associated with stronger purchase conversion.
- Review low- and zero-conversion traffic sources to identify opportunities for traffic-quality improvement or more targeted acquisition strategies.

Purchase sessions generally show higher product-related page engagement than non-purchase sessions.



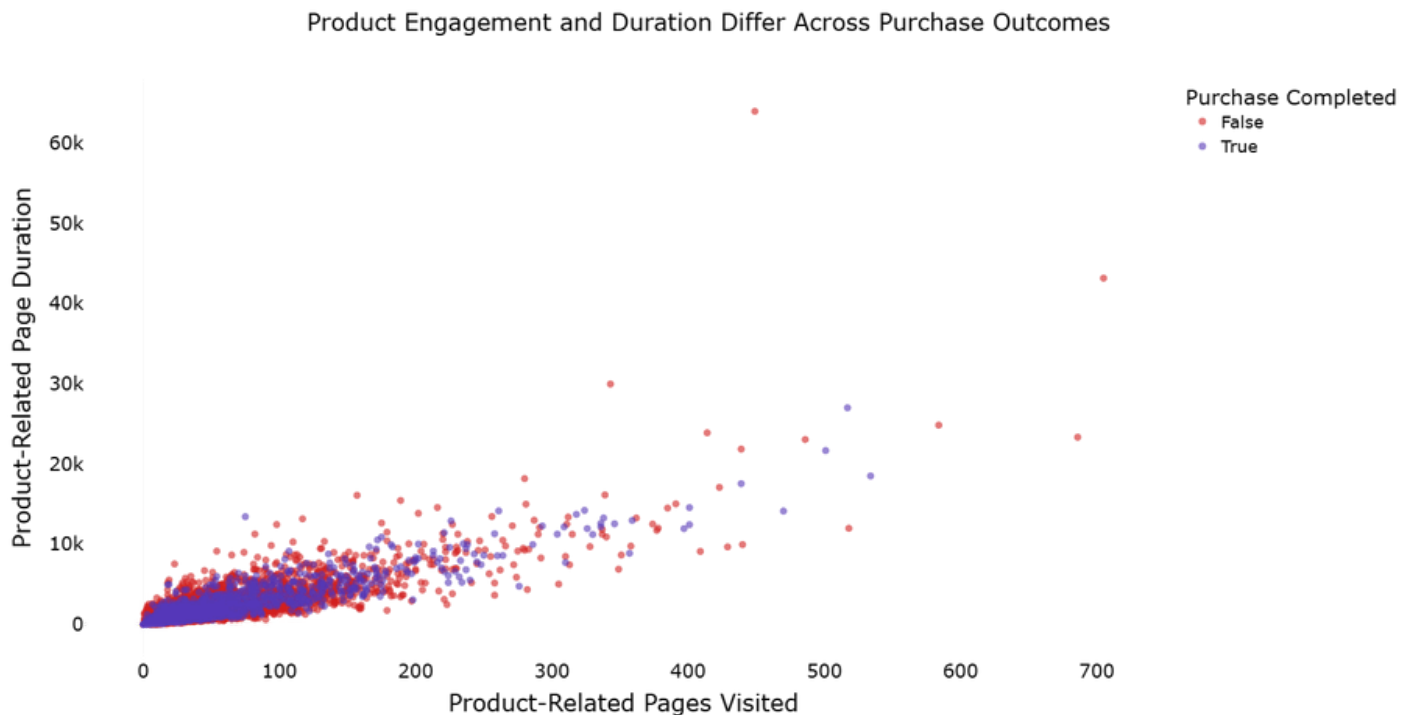
Key observations

- The median number of ProductRelated pages is higher for purchase sessions than for non-purchase sessions.
- Both groups contain substantial high-engagement outliers, with non-purchase sessions reaching above 700 product-related page visits and purchase sessions reaching above 500.

Business Insights

- Product-page engagement can be monitored as an important behavioral signal when identifying sessions with stronger purchase intent.
- High product-page activity does not guarantee conversion, so additional behavioral signals should be considered when distinguishing purchasers from highly engaged non-purchasers.

Product-related engagement & browsing duration show a positive relationship, with purchase and non-purchase sessions distributed across the same overall engagement range.



Key observations

- Product-related pages visited and product-related page duration increase together, indicating that sessions with more product-page activity generally spend more time on those pages.
- Purchase and non-purchase sessions substantially overlap across the engagement range, although purchase sessions are visible throughout higher-engagement regions.

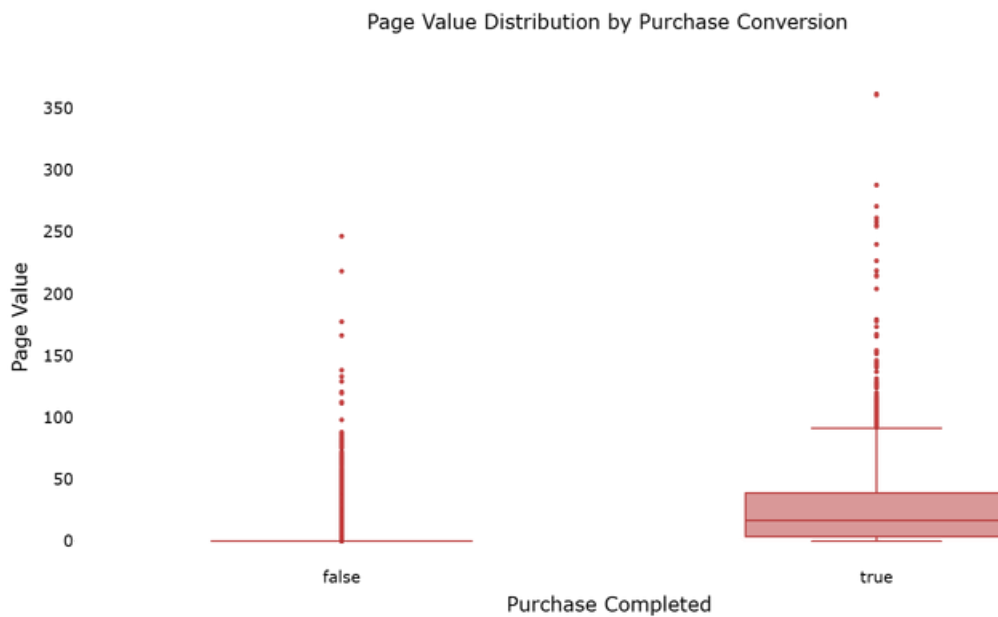
Business Insights

- Combine product-page engagement and browsing duration with other conversion signals when assessing session-level purchase intent.
- Investigate high-engagement non-purchase sessions to understand where substantial product interest does not translate into a recorded purchase.

Purchase sessions show substantially higher Page Value levels than non-purchase sessions

52.03%

Typical CPU Utilization (Median)



Key observations

- The distribution for purchase sessions is clearly shifted upward, with a much higher central range than non-purchase sessions.
- Non-purchase sessions are heavily concentrated near zero Page Value, while purchase sessions extend across substantially higher values.

Business Insights

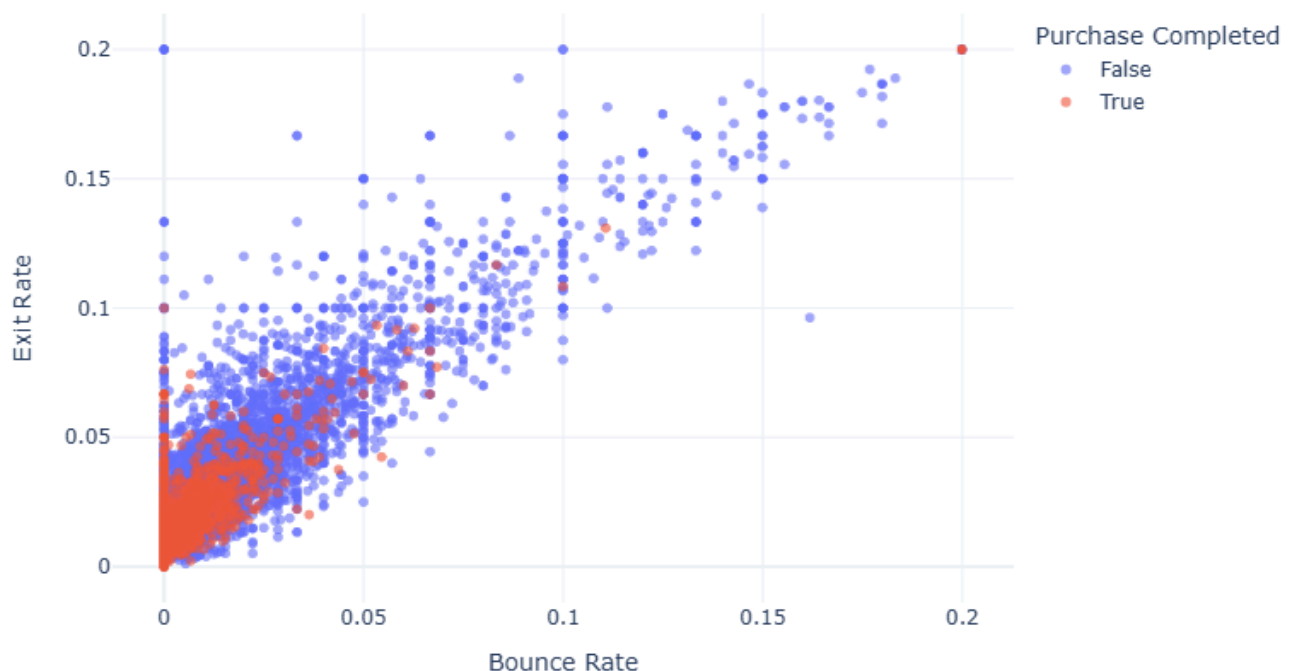
- Page Value can serve as a strong behavioral signal when identifying sessions with stronger purchase potential.
- Monitor Page Value alongside other session characteristics rather than treating it as a standalone indicator of conversion.

Higher discounts do not consistently translate into higher demand, indicating a more complex pricing relationship

51.63%

Typical Memory Utilization (Median)

Bounce and Exit Behavior Differ Across Purchase Outcomes



Key observations

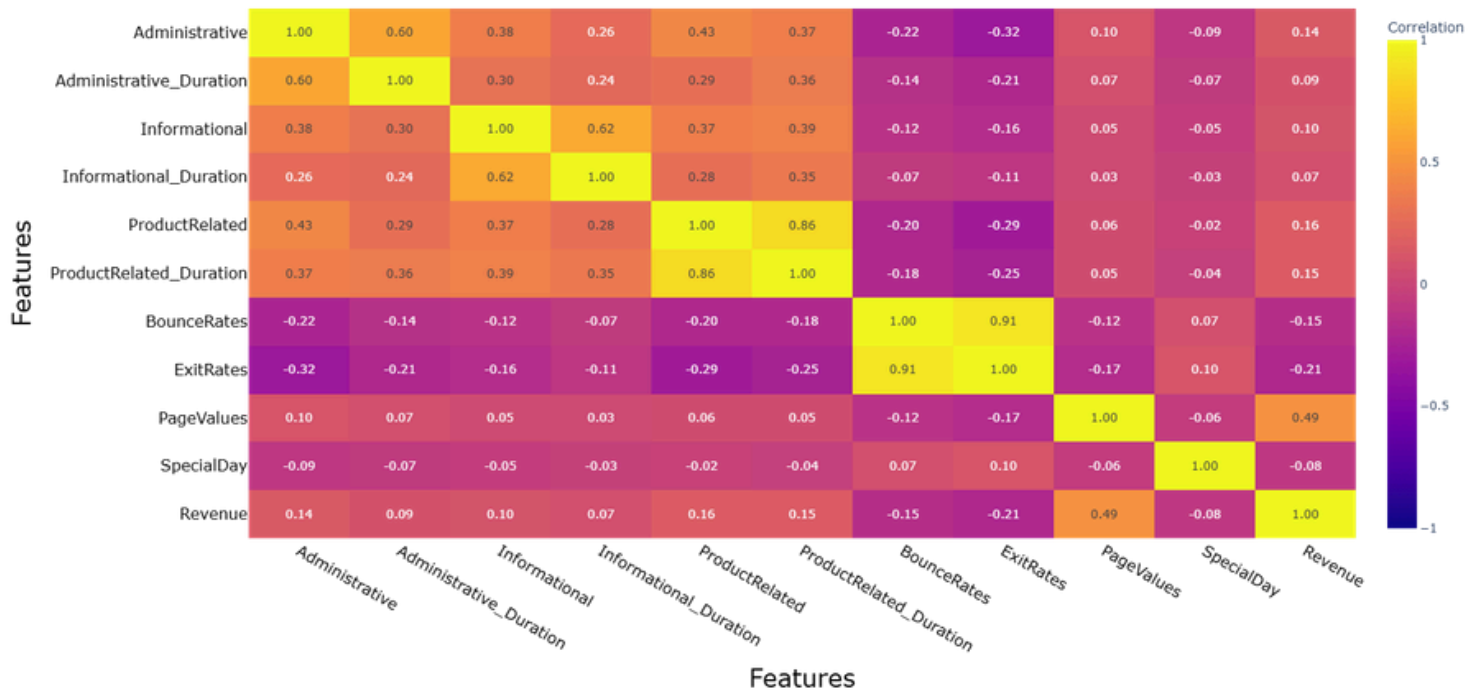
- Demand varies substantially across every discount level.
- There is no obvious simple linear relationship where increasing discount consistently produces increasing demand.

Business Insights

- Larger discounts do not automatically guarantee proportionally higher demand.
- Discount strategies should therefore be evaluated alongside product characteristics, inventory, promotions, seasonality, and other demand drivers.

Sunny conditions are associated with higher average demand, while adverse weather corresponds to lower demand

Numerical Features Show Varying Relationships with Purchase Conversion



Key observations

- Sunny conditions have the highest average demand, at approximately 115 units.
- Rainy and Snowy conditions have lower average demand, around 94–95 units.

Business Insights

- Weather conditions can contribute to differences in retail demand and should therefore be considered during forecasting.
- Businesses may benefit from adjusting inventory planning when weather patterns indicate potential changes in demand.